ELEC 457 PLL Project

Student Name and Number: Thomas Wilkinson, 20269155

Student Name and Number: Adam Hazenberg, 20293978

Question 1

For the PLL project, the are 5 parts. The project is out of 20.

1. [5] For a 15% overshoot and desired change in error = 18° for a shift from 1.0 MHz to 1.2 MHz as well as desired change in error = 45° for a shift from 1.0 MHz to 1.5 MHz: Calculate your desired damping factor ζ, VCO constant Ko (in $\frac{(rad)}{V}$), lower cut off frequency f_L, ω_L and component values. You will have to choose a phase detector to determine your Kd, but assume you are using a basic RC Low Pass Filter and have V_{DD} = 10 V.

Calculating VCO constant K_o:

Choosing an XOR phase detector:

$$K_d = \frac{V_{DD}}{\pi} \frac{V}{rad} = \frac{10}{\pi} \frac{V}{rad}$$

$$\Delta f_1 = 1.2MHz - 1.0MHz = 0.2\ MHz \rightarrow \Delta w_1 = 0.4\pi \times 10^6$$

$$\Delta f_2 = 1.5MHz - 1.0MHz = 0.5\ MHz \rightarrow \Delta w_2 = \pi \times 10^6$$

$$\Delta \phi_{e,1} = 18^\circ \times \frac{\pi}{180^\circ} = 0.31416\ rad$$

$$\Delta \phi_{e,2} = 45^\circ \times \frac{\pi}{180^\circ} = \frac{\pi}{4}\ rad$$

$$K_{o_1} = \frac{\Delta w}{K_d \Delta \phi_{e,1} F(s)} = \frac{0.4\pi \times 10^6 \frac{1}{s}}{\left(\frac{10}{\pi}\frac{V}{rad}\right)(0.31416\ rad)(1)} = 1.257\ M \frac{rad}{Vs}$$

$$K_{o_2} = \frac{\Delta w}{K_d \Delta \phi_{e,2} F(s)} = \frac{\pi \times 10^6 \frac{1}{s}}{\left(\frac{10}{\pi}\frac{V}{rad}\right)\left(\frac{\pi}{4} rad\right)(1)} = 1.257\ M \frac{rad}{Vs}$$

$$K_o = 1.257\ M \frac{rad}{Vs} \rightarrow K_o = \frac{1.257\ M\ rad/Vs}{2\pi} = 200057.8 = 200\ kHz/V$$

Calculating f_L, w_L and RC:

Basic RC low Pass: $F(s) = \frac{1}{1+sRC}$

$$s^2 + \frac{s}{RC} + \frac{K_d K_o}{RC} = 0$$

$$s^2 + 2\zeta w_n s + w_n^2 = 0$$

$$w_n = \sqrt{\frac{K_d K_o}{RC}} \qquad 2\zeta w_n = \frac{1}{RC} \rightarrow w_n = \frac{1}{2\zeta RC}$$

$$\sqrt{\frac{K_d K_o}{RC}} = \frac{1}{2\zeta RC}$$

$$2\zeta\sqrt{K_d K_o} = \frac{\sqrt{RC}}{RC}$$

$$4\zeta^2 K_d K_o = \frac{1}{RC} \rightarrow RC = \frac{1}{4(0.5169)^2\left(\frac{10}{\pi}\right)(1.257 \times 10^6)} = 2.3385 \times 10^{-7}$$

$$w_n = \frac{1}{2\zeta RC} = \frac{1}{2(0.5169)(2.3385 \times 10^{-7})} = 4.136 \times 10^6\ rad/s$$

$$w_L = 2\zeta w_n = 2(0.5169)(4.135 \times 10^6) = 4.276 \times 10^6\ rad/s$$

$$f_L = \frac{w_L}{2\pi} = \frac{4.27 \times 10^6}{2\pi} = 680579.075\ Hz$$

$RC = 2.3385 \times 10^{-7}\ s$, choosing $C = 4nF$

with $C = 4nF \rightarrow R = \frac{2.3385 \times 10^{-7}}{4 \times 10^{-9}} = 58.46\ \Omega$

Calculating ζ:

$$15\% = 100\, e^{\frac{-\zeta\pi}{\sqrt{1-\zeta^2}}}$$

$$\ln(0.15) = \frac{-\zeta\pi}{\sqrt{1-\zeta^2}}$$

$$\ln^2(0.15)(1-\zeta^2) = \zeta^2\pi^2$$

$$\ln^2(0.15) = \zeta^2(\ln^2(0.15) + \pi^2)$$

$$\zeta = \sqrt{\frac{\ln^2(0.15)}{\ln^2(0.15) + \pi^2}} = 0.5169$$

Figure 1: Calculations for part 1.

Question 2

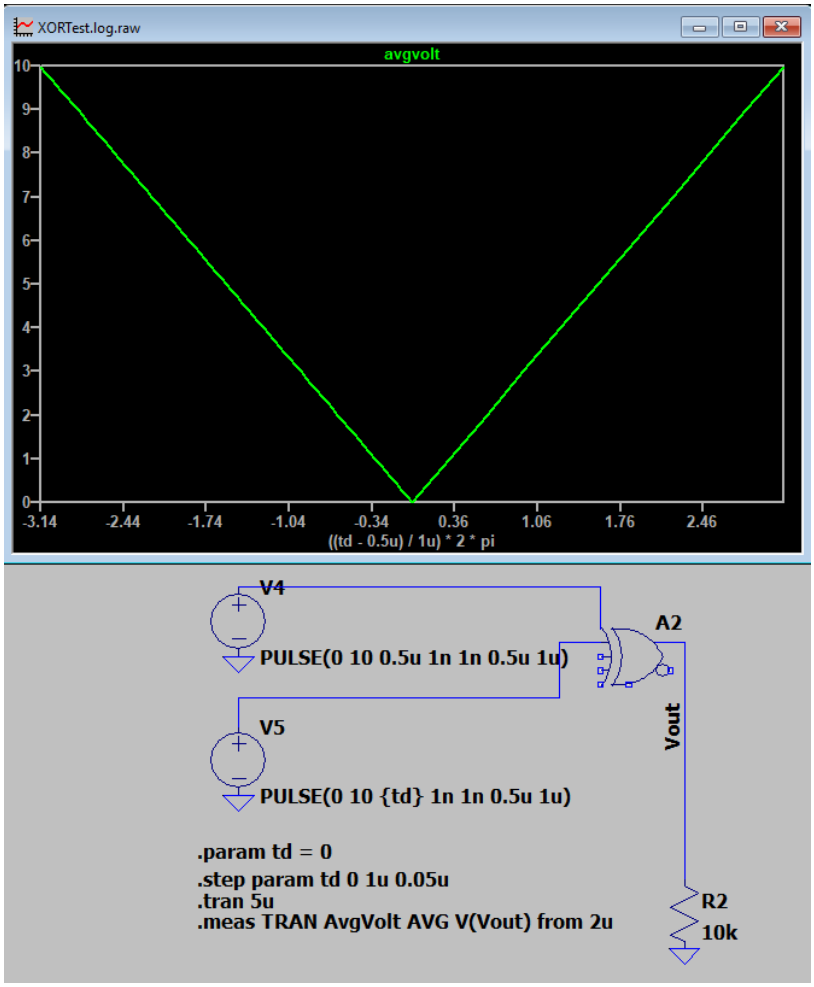

Figure 2: Plot of average output voltage versus phase error from -3.14 to 3.14 rad.

The phase detector used was a standard XOR gate with a Vhigh of 10V and a threshold of 0.5V. To simulate -pi to pi phase shift, two 1MHz pulse waves were input into the XOR gate, one simulating the reference signal (V4) and one simulating the VCO signal (V5). The reference frequency was kept constant and the VCO's phase was varied around it. This was done by always starting the reference pulse at 0.5us delay and continuously varying the start of the VCO from 0s start delay (-pi phase shift) to 1us start delay (pi phase shift). The average voltage output vs phase error was calculated using the AvgVolt measuring tool in LTSpice and can be seen plotted in Figure 2 where the y axis is the output voltage of the XOR gate, and the x axis is the phase shift in radians between the two input signals. To simulate the output for 18-degree and 45-degree phase shift, the reference signals start delay was kept at 0us. Then the VCOs start delay was set to 50ns for 18-degree phase shift and 125ns for 45-degree phase shift. The calculations and XOR output plots for the time delays for each phase error are shown below.

$\theta_e \approx 18°$:

$$t_{delay} = T\frac{18°}{360°} = 1us \ \times 0.05 = 50ns$$

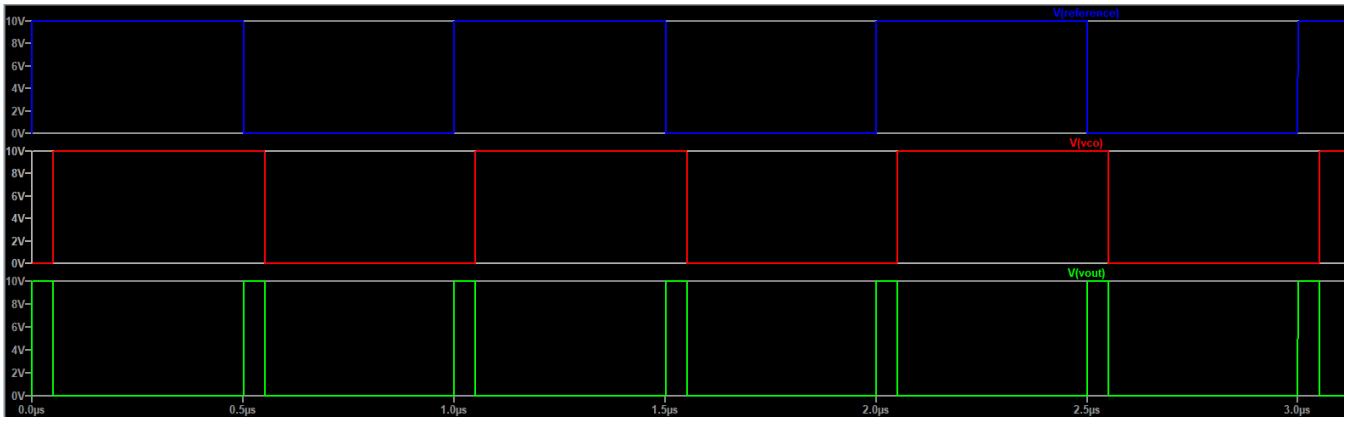

Figure 3: Phase detector output for phase shift of 18 degrees.

$\theta_e \approx 45°$:

$$t_{delay} = T\frac{45°}{360°} = 1us \ \times 0.125 = 125ns$$

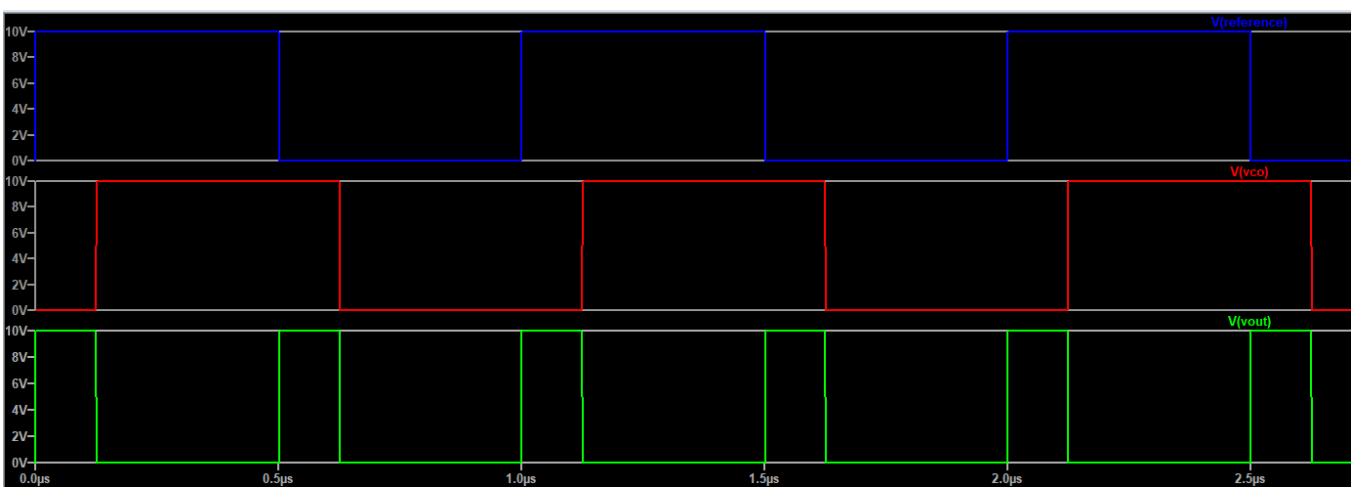

Figure 4: Phase detector output for phase shift of 45 degrees.

Question 3

Initial RC filter:

$$R = 58.46\Omega,\ C = 4nF,\ RC = 2.3385 \times 10^{-7}s$$

$$f_L = \frac{1}{2\pi RC} = 680579.075\ Hz$$

$$H(s) = \frac{V_c(s)}{V_p(s)} = \frac{1}{1 + sRC} = \frac{1}{1 + s(2.3385 \times 10^{-7})}$$

$$\frac{V_c}{V_p}(f) = \frac{1}{1 + j2\pi fRC} = \frac{1}{1 + j2\pi f(2.3385 \times 10^{-7})} = \frac{1}{1 + jf(1.469337 \times 10^{-6})}$$

$$\frac{V_c}{V_p}(f) = \frac{1}{1 + j\frac{f}{680579}}$$

Final RC filter:

$$R = 600\Omega,\ C = 4nF,\ RC = 2.4 \times 10^{-6}s$$

$$f_L = \frac{1}{2\pi RC} = 66314.56\ Hz$$

$$H(s) = \frac{V_c(s)}{V_p(s)} = \frac{1}{1 + sRC} = \frac{1}{1 + s(2.4 \times 10^{-6})}$$

$$\frac{V_c}{V_p}(f) = \frac{1}{1 + j2\pi fRC} = \frac{1}{1 + j2\pi f(2.4 \times 10^{-6})} = \frac{1}{1 + jf(1.508 \times 10^{-5})}$$

$$\frac{V_c}{V_p}(f) = \frac{1}{1 + j\frac{f}{66314.56}}$$

Question 4

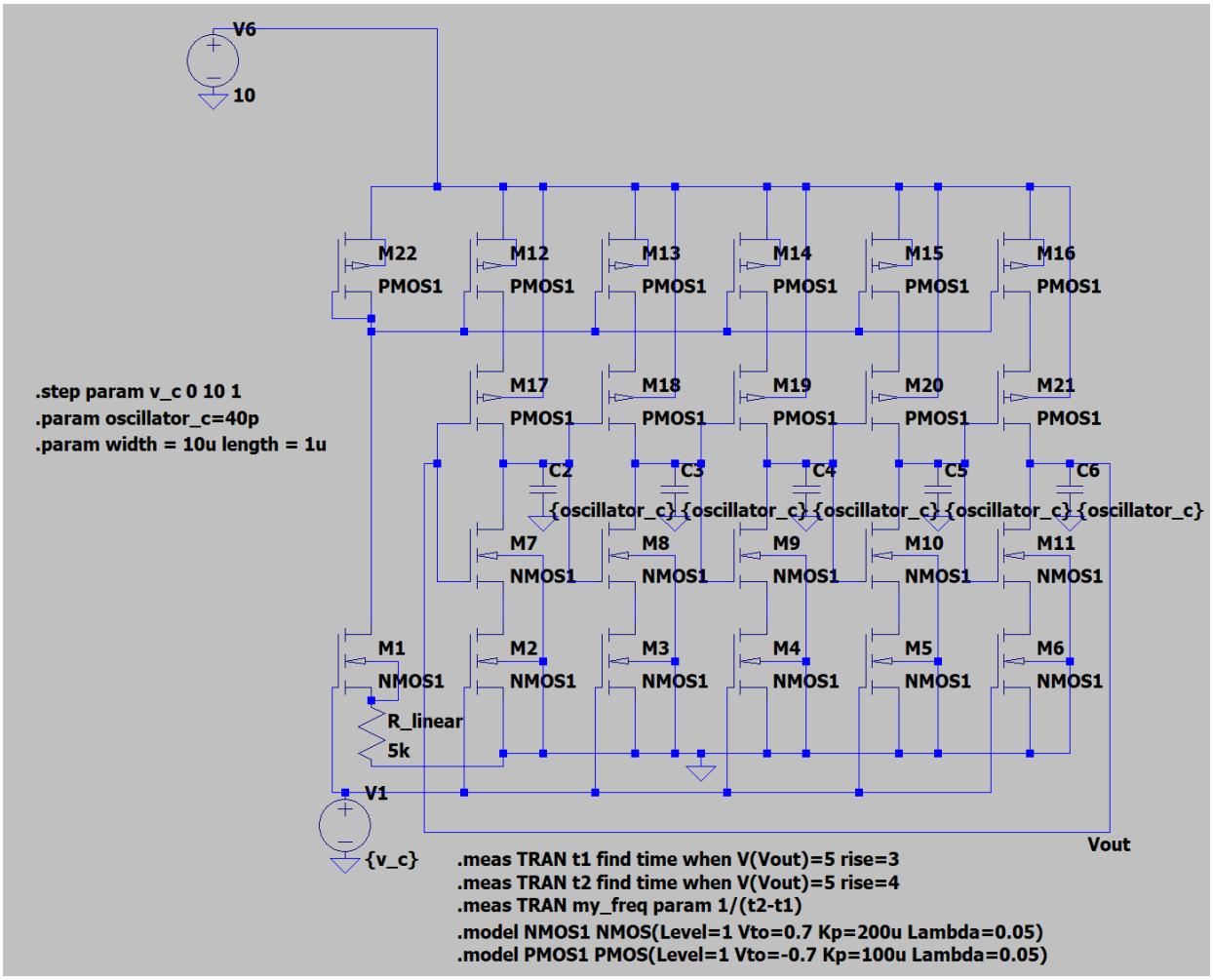

Figure 5: 5 inverter ring oscillator used as the VCO.

Table 1:Table showing values of control voltage vs output frequency of the VCO.

Control Voltage (Volts)	Output Frequency (Hz)
0	0
1	0
2	459875
3	765890
4	1.03E+06
5	1.26E+06
6	1.47E+06
7	1.67E+06
8	1.86E+06
9	2.02E+06
10	2.10E+06

As seen in Figure 6, the team tweaked oscillator_c, R_linear, and the widths and lengths of the MOSFETs until the optimal Ko curve was achieved. The team was looking to get the 1-1.5 MHz range linear and have the slope in that range be as close as possible to the initially calculated Ko (200kHz/v). As seen in Figure 5, the optimal values were a capacitor sizes of 40pF, resistor size of 5kΩ, and MOSFET width and lengths of 10um and 1um.

Calculating Ko from Table 1 values:

$$K_o = \frac{1.67 \times 10^6 - 1.03 \times 10^6}{7 - 4} = 213\,333\ Hz/V$$

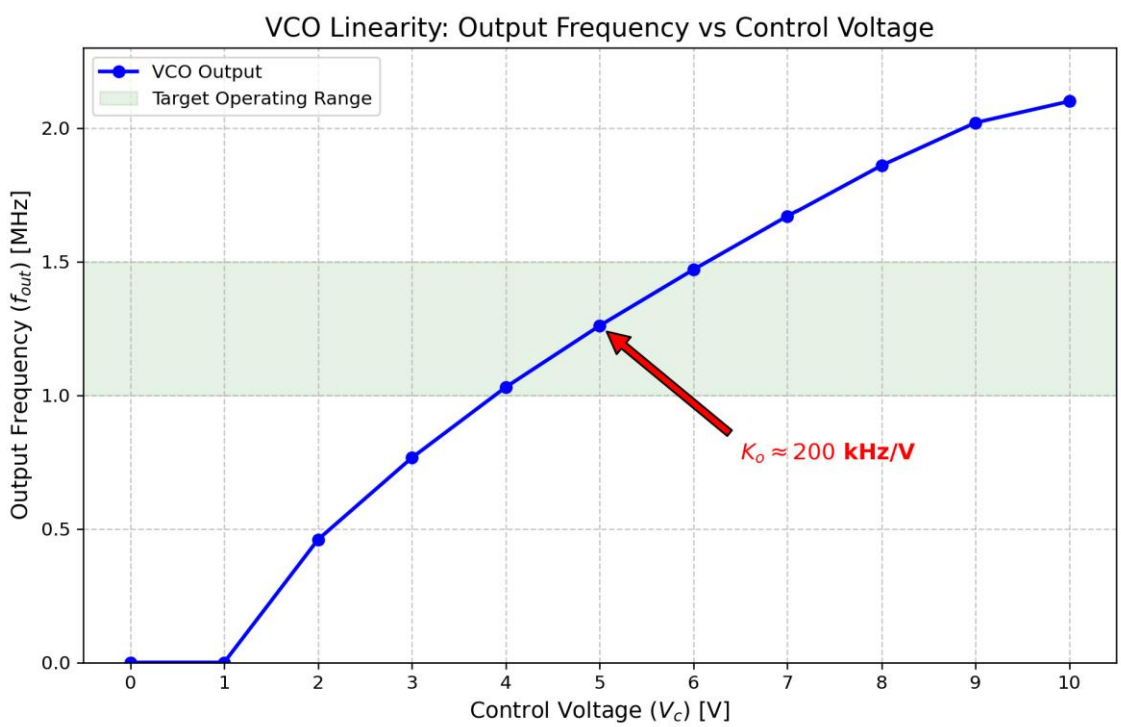

Figure 6: Plot depicting output frequency vs control voltage for the VCO. (using values from Table 1)

Question 5

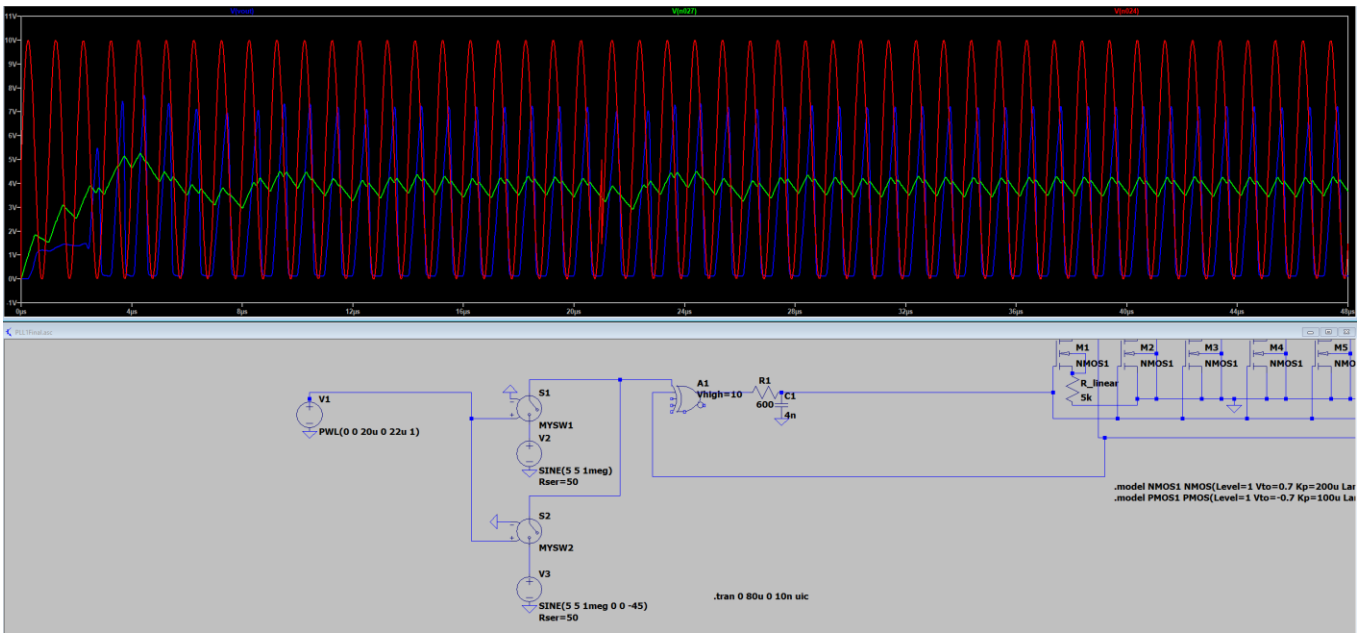

Figure 7: PLL's response to -pi/4 phase shift. Red signal is V_ref (reference input), blue signal is V_out (VCO's output), and green signal is V_c (control voltage at the output of the RC filter).

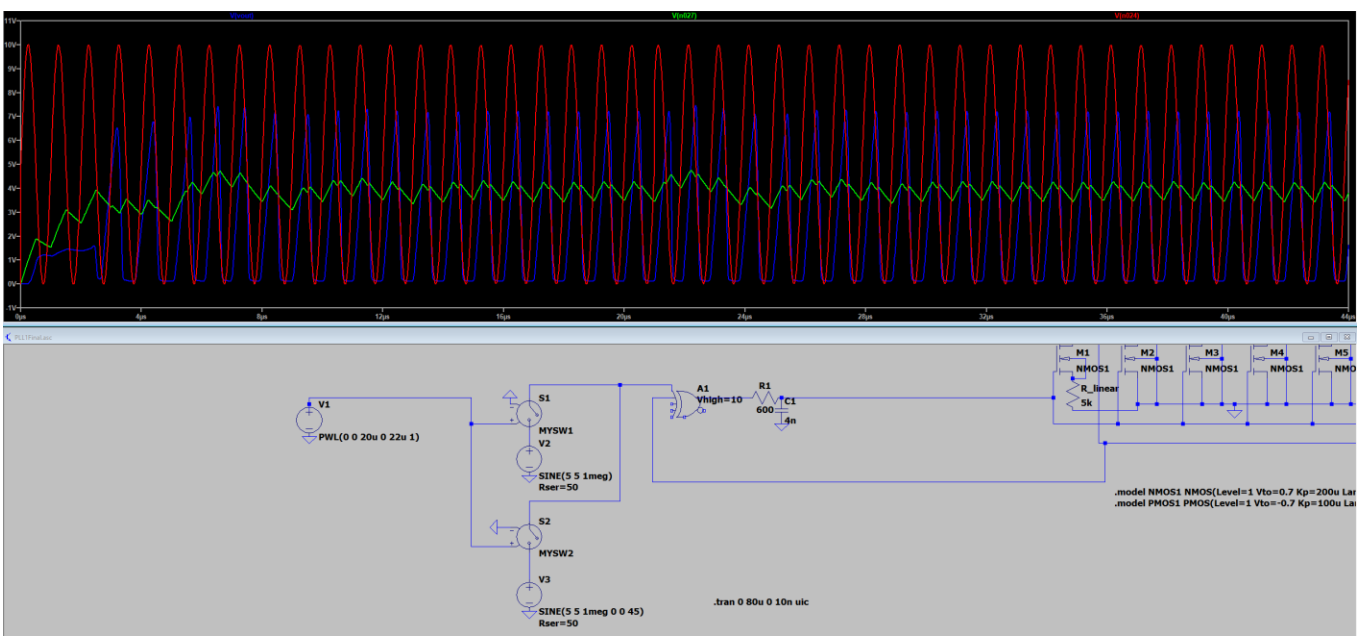

Figure 8: PLL's response to pi/4 phase shift. Red signal is V_ref (reference input), blue signal is V_out (VCO's output), and green signal is V_c (control voltage at the output of the RC filter).

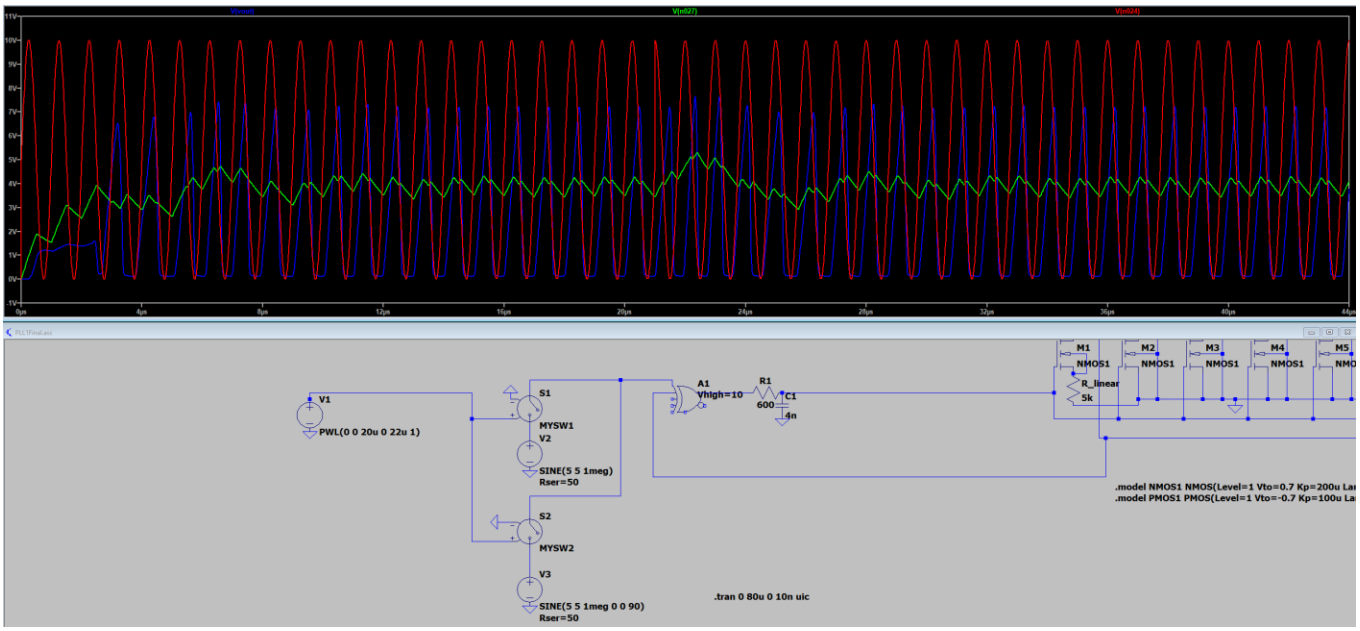

Figure 9: PLL's response to pi/2 phase shift. Red signal is V_ref (reference input), blue signal is V_out (VCO's output), and green signal is V_c (control voltage at the output of the RC filter).

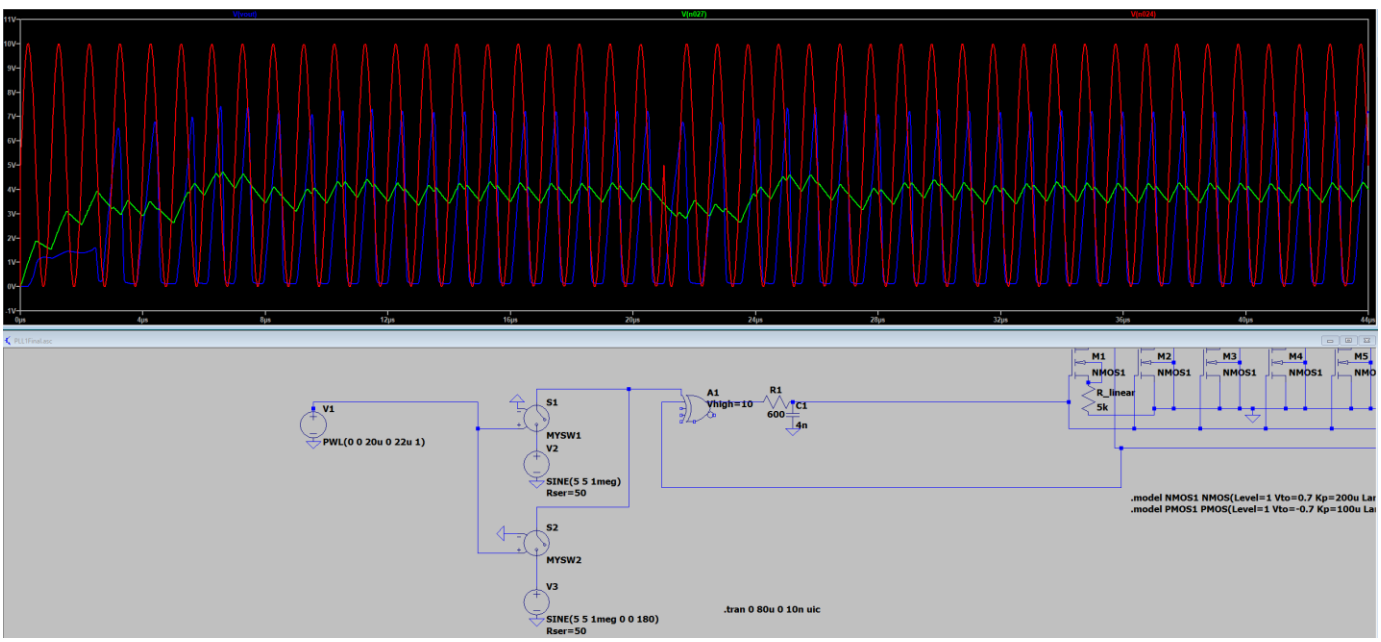

Figure 10: PLL's response to pi phase shift. Red signal is V_ref (reference input), blue signal is V_out (VCO's output), and green signal is V_c (control voltage at the output of the RC filter).

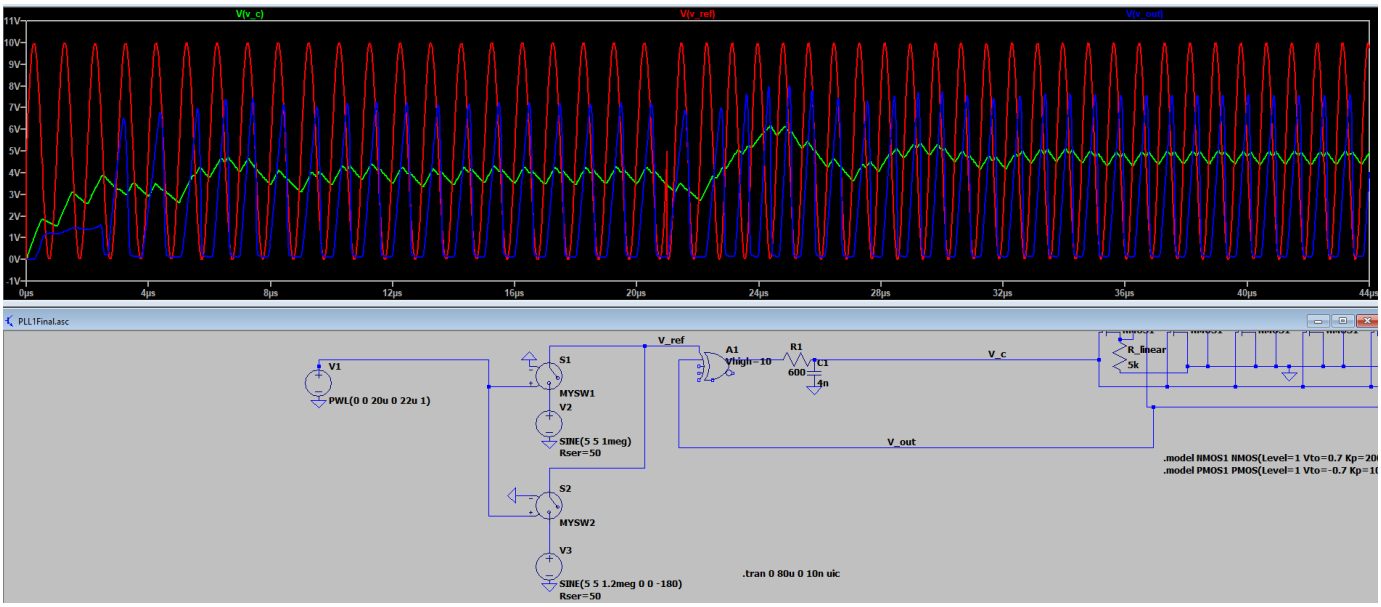

Figure 11: PLL's response to input frequency shift of 1MHz to 1.2MHz. Red signal is V_ref (reference input), blue signal is V_out (VCO's output), and green signal is V_c (control voltage at the output of the RC filter).

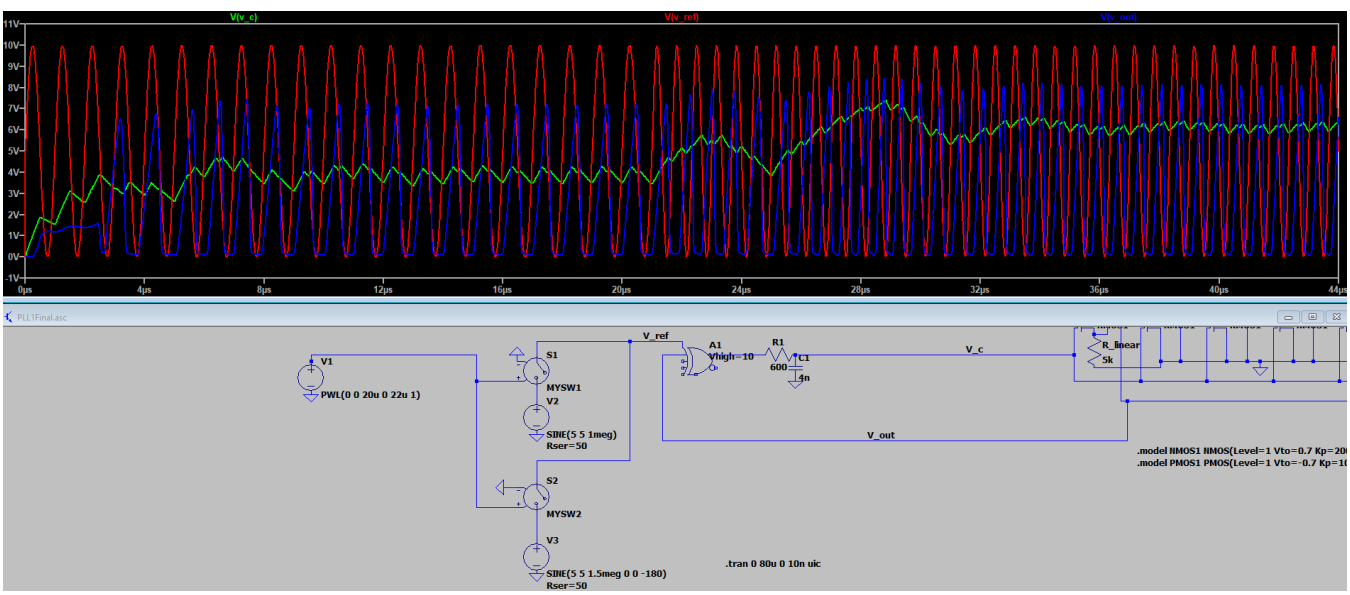

Figure 12: PLL's response to frequency shift of 1MHz to 1.5MHz. Red signal is V_ref (reference input), blue signal is V_out (VCO's output), and green signal is V_c (control voltage at the output of the RC filter).

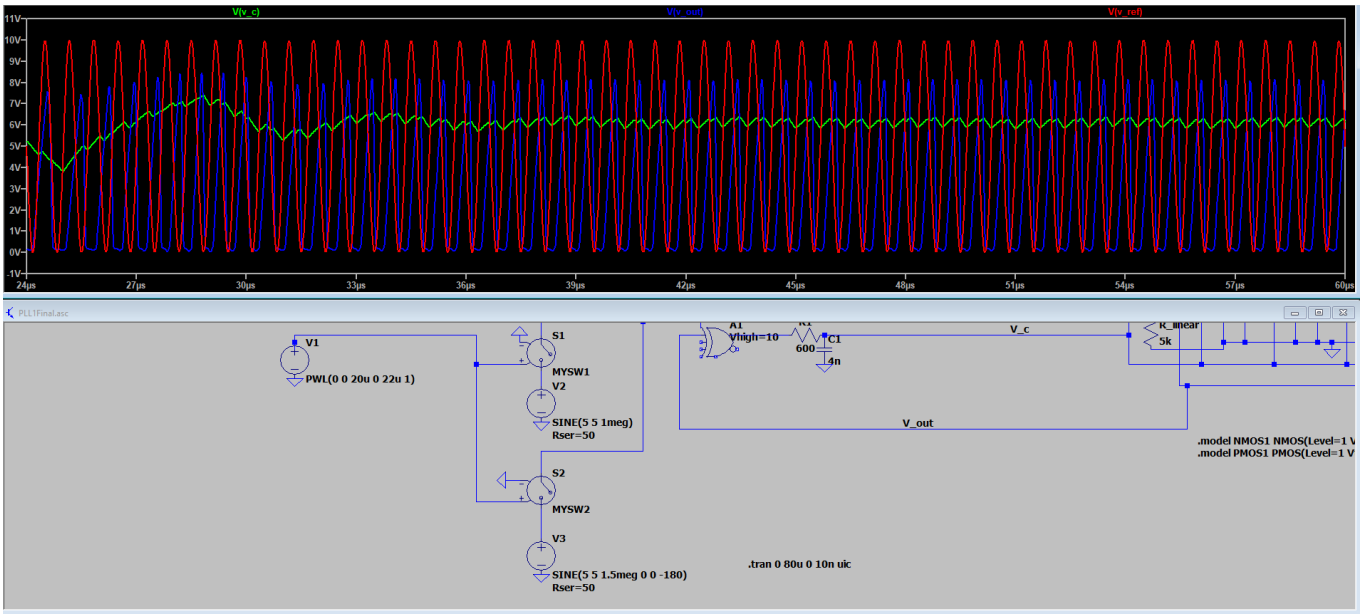

Figure 13: PLL's response to frequency shift of 1MHz to 1.5MHz continued.

Actual Values:

$$K_d = \frac{V_{DD}}{\pi} = \frac{10}{\pi} V/rad$$

(Calculated in Q4) $K_o = 213\,333.333 \frac{Hz}{V} \times 2\pi = 1.340413 \times 10^6\ rad/Vs$

$$RC = (600)(4 \times 10^{-9}) = 2.4 \times 10^{-6} s$$

$$w_n = \sqrt{\frac{K_d K_o}{RC}} = \sqrt{\frac{\frac{10}{\pi}(1.340413 \times 10^6)}{2.4 \times 10^{-6}}} = 1.3333 \times 10^6\ rad/s$$

$$\zeta = \frac{1}{2 w_n RC} = \frac{1}{2(1.3333 \times 10^6)(2.4 \times 10^{-6})} = 0.156$$

$$w_L = 2\zeta w_n = 2(0.156)(1.3333 \times 10^6) = 416665.6\ rad/s$$

Zeta from actual simulated PO (using V_c plot from Figure 12 to approximate):

$$PO \approx \frac{Max\ Peak - Settle\ Voltage}{Settle\ Voltage - Initial\ Voltage} \times 100\% \approx \frac{7.4 - 6.36}{6.36 - 4.26} \times 100\% \approx 0.495 \times 100\% \approx 49.5\%$$

$$\zeta \approx \sqrt{\left(\frac{\ln(0.495)^2}{\ln(0.495)^2 + \pi^2}\right)} \approx 0.2183$$